

Numerical Results

Tables and figures for all three experiments,
the sensitivity studies, and the clamping-boundary study

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Part I

Main results

1 Experiment 1: clamped location-scale normal

$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} N(\mu^*, \sigma^{*2})$ clamped to $[L, U] = [0, 3]$; the release is the clamped mean and clamped sample variance with Gaussian noise of scale $(U - L)/(n\varepsilon)$ and $(U - L)^2/(n\varepsilon)$. Throughout, $n = 100$, $\varepsilon = 1$, $\mu^* = \sigma^* = 1$, $R = 200$, $R_{\text{aux}} = 400$, $\lambda_n = 1/\log n = 0.2171$, $\alpha = 0.05$, and 1000 Monte Carlo replications. The finite- R coverage bound is $(R + 2 - a_{R,\alpha})/(R + 1) = 0.95025$ with $a_{R,\alpha} = 11$; the Monte Carlo standard error at that level is 0.0069.

Table 1: Experiment 1. Empirical coverage and average width of the nominal 95% intervals, and average area of the joint region. Monte Carlo standard errors in parentheses. Joint coverage is obtained by evaluating the acceptance rule at the true parameter, so it involves no nuisance optimisation and is a direct check on the implemented statistic.

Method		Marginal		Joint	Area
		μ	σ		
Mahalanobis	Coverage	0.991 (0.003)	0.985 (0.004)	0.963 (0.006)	0.341 (0.004)
	Width	0.600 (0.003)	0.758 (0.004)		
Penalized Wald (μ)	Coverage	0.966 (0.006)	1.000 (0.000)	0.956 (0.006)	0.657 (0.007)
	Width	0.504 (0.003)	1.730 (0.009)		
Penalized Wald (σ)	Coverage	1.000 (0.000)	0.964 (0.006)	0.956 (0.006)	0.503 (0.005)
	Width	1.076 (0.005)	0.614 (0.003)		
PB-ADI	Coverage	0.959 (0.006)	0.960 (0.006)	0.967 (0.006)	0.338 (0.004)
	Width	0.463 (0.003)	0.576 (0.003)		

Notes: the two Penalized Wald rows are separate runs, each orthogonalised against the other coordinate; the off-target column reports the price of that orthogonalisation and the pair does not carry simultaneous coverage. Area is reported for completeness but is not the relevant criterion for a targeted statistic. Numerical failures were zero for all four methods, and no interval was returned at the search-box boundary in any replication.

Table 2: Experiment 1. Width reduction of the targeted Penalized Wald interval relative to the Mahalanobis interval for the same parameter, with the coverage of the targeted interval. The finite- R coverage bound is 0.95025.

Target	Mahalanobis width	Penalized Wald width	Reduction	Coverage
μ	0.600	0.504	15.9%	0.966 (0.006)
σ	0.758	0.614	19.0%	0.964 (0.006)

Sensitivity of Experiment 1 to the penalty weight λ_n , the inference cloud size R and the auxiliary cloud size R_{aux} is reported in Part II, together with the clamping and privacy sweeps of the other two experiments.

2 Experiment 2: simple linear regression, $H_0 : \beta_1 = 0$

$Y = \beta_0^* + X\beta_1^* + \epsilon$ with $X \sim N(0.5, 1)$, $\epsilon \sim N(0, 0.25)$, $\beta_0^* = -0.5$, sufficient-statistic perturbation of five clamped moments at $\Delta = 2$ under 1-GDP, $R = 200$, $\alpha = 0.05$, 1000 replications per cell. The two comparison panels are the published grids: REPRO is the “Repro Sample” panel of Awan and Wang (2025, Fig. 5) and PB-ADI is the “PB (Indirect estimator + approximate-pivot)” panel of Wang, Chang and Awan (Fig. 5), both restricted to the cells of our design. The exact finite- R size bound is $(a_{R,\alpha} - 1)/(R + 1) = 0.04975$.

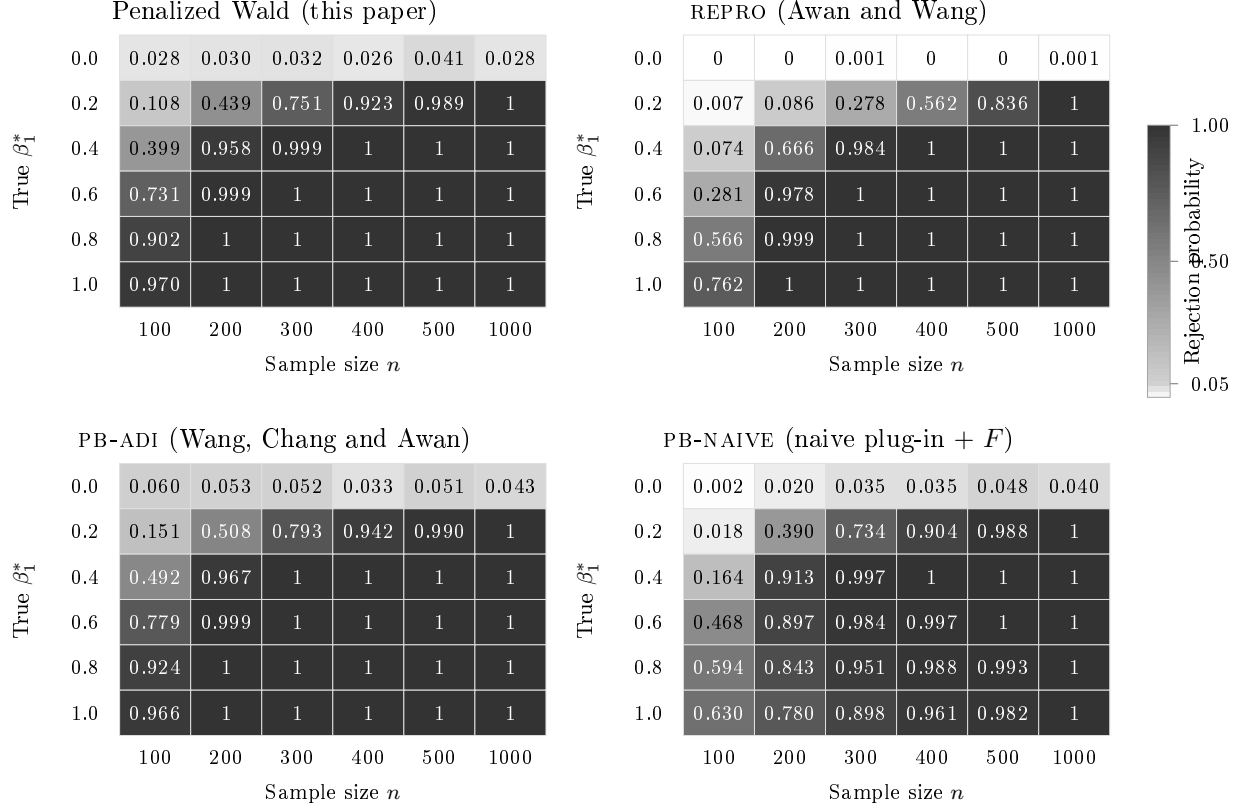


Figure 1: Experiment 2. Rejection probability of $H_0 : \beta_1 = 0$ at the 0.05 significance level. Rows are the true slope β_1^* , columns the sample size n . The grey scale runs from 0 to 1 and breaks at 0.05, so cells at or below the nominal level are light and cells above it are dark: the bottom row of each panel is empirical size and the rest is power. Our size is 0.026–0.041, at or below the exact bound 0.04975 in every cell, while PB-ADI reaches 0.060 at $n = 100$ and PB-NAIVE rises with n . REPRO has size 0.000–0.001 and pays for it in power.

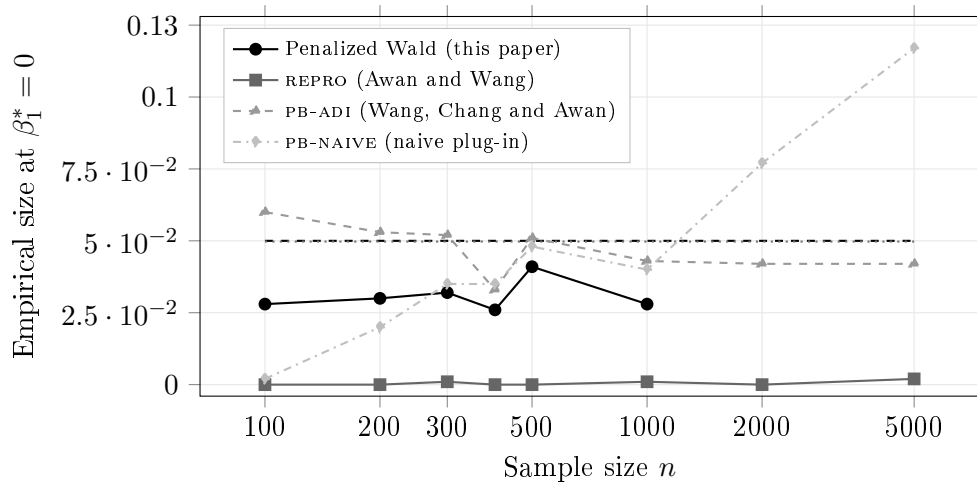


Figure 2: Experiment 2. Empirical size at $\beta_1^* = 0$ against sample size. The dashed line is the nominal level 0.05 and the dotted line the exact finite- R bound 0.04975. Our test stays below the bound at every n it was run at; PB-NAIVE deteriorates as n grows, reaching 0.117 at $n = 5000$; PB-ADI sits at or slightly above the nominal level; REPRO is essentially zero. Our design stops at $n = 1000$, so the two largest sample sizes are shown for the published methods only.

3 Experiment 3: logistic regression via objective perturbation

$x_i = 2z_i - 1$ with $z_i \sim \text{Beta}(a^*, b^*)$ and $y_i \mid x_i \sim \text{Bern}(\text{expit}(\beta_0^* + \beta_1^* x_i))$, $a^* = b^* = 1$, $\beta_0^* = 0.5$, $\beta_1^* = 2$; the release combines ℓ_∞ objective perturbation for (β_0, β_1) with the optimal K -norm mechanism for the first two moments of z , under ε -DP. $R = 200$, 1000 replications per cell, and all intervals are at 90% confidence, so the finite- R coverage bound is $(R + 2 - a_{R,\alpha})/(R + 1) = 0.90050$ with $a_{R,\alpha} = 21$; the Monte Carlo standard error at that level is 0.0095. Comparison panels are the “Repro Samples”, “Debiased Estimator” and “Naive Estimator” panels of Wang, Chang and Awan (Fig. 6), restricted to the cells of our design.

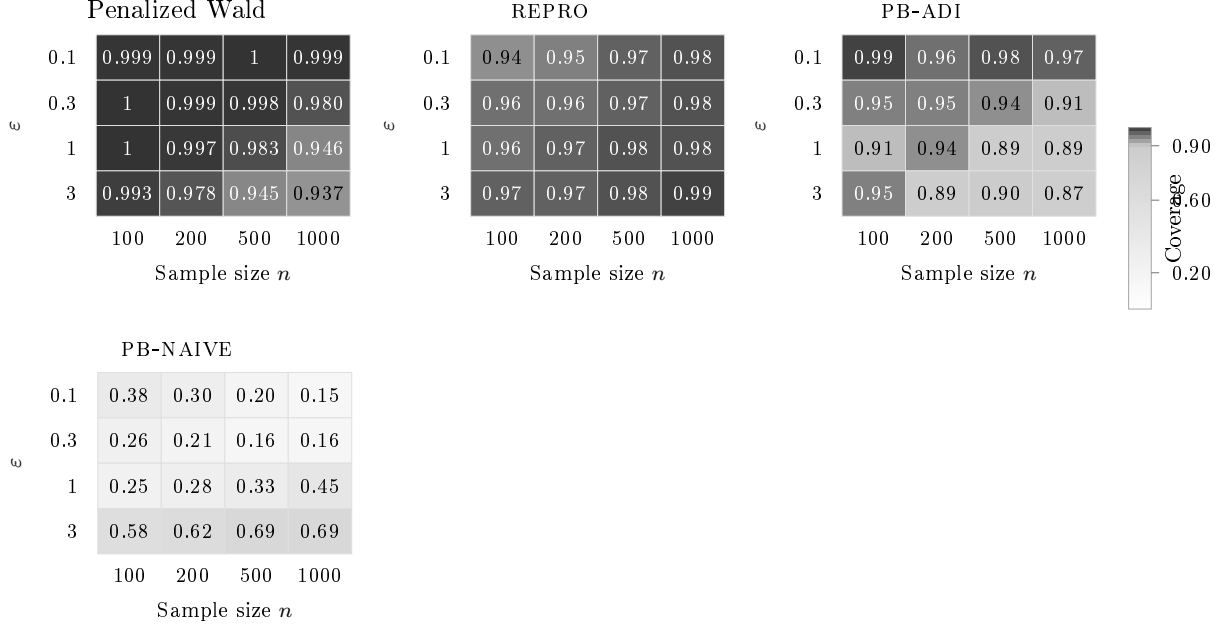


Figure 3: Experiment 3. Empirical coverage of the 90% interval for β_1^* . Rows are the privacy parameter ε , columns the sample size n . The grey scale runs from 0 to 1 and breaks at the nominal level 0.90, so cells below it are light. PB-NAIVE under-covers everywhere, by an order of magnitude at small ε . Our procedure and REPRO over-cover throughout; PB-ADI sits closest to the nominal level and dips slightly below it in three cells.

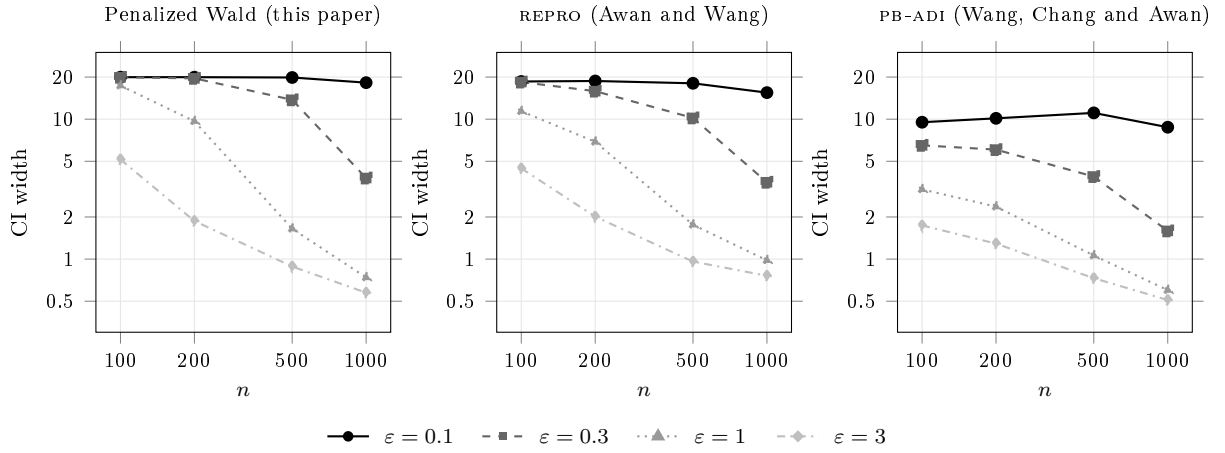


Figure 4: Experiment 3. Average width of the 90% interval for β_1^* , on logarithmic axes, one curve per privacy parameter. The search box is $[-10, 10]$, so a width near 20 means the interval fills the box and is uninformative: this is the case for $\epsilon = 0.1$ at every sample size, for both our procedure and REPRO, and those cells carry no information about relative efficiency. In the informative regime ($\epsilon \geq 1, n \geq 500$) our widths are 6–25% below REPRO and remain above PB-ADI. PB-NAIVE is omitted: its widths are the smallest of all four methods, which is a consequence of its under-coverage in Figure 3 rather than of efficiency.

Table 3: Experiment 3. Coverage and average width of the 90% interval for β_1^* in the cells common to all four methods. Values for REPRO, PB-ADI and PB-NAIVE are read from Wang, Chang and Awan (Fig. 6).

ε	n	Coverage				Width		
		PW	REPRO	PB-ADI	PB-NAIVE	PW	REPRO	PB-ADI
0.1	100	0.999	0.94	0.99	0.38	19.967	18.59	9.51
	200	0.999	0.95	0.96	0.30	19.963	18.75	10.14
	500	1.000	0.97	0.98	0.20	19.860	18.05	11.09
	1000	0.999	0.98	0.97	0.15	18.243	15.48	8.76
0.3	100	1.000	0.96	0.95	0.26	19.912	18.42	6.50
	200	0.999	0.96	0.95	0.21	19.585	15.86	6.06
	500	0.998	0.97	0.94	0.16	13.751	10.21	3.89
	1000	0.980	0.98	0.91	0.16	3.796	3.54	1.59
1	100	1.000	0.96	0.91	0.25	17.366	11.40	3.15
	200	0.997	0.97	0.94	0.28	9.658	6.89	2.37
	500	0.983	0.98	0.89	0.33	1.648	1.76	1.06
	1000	0.946	0.98	0.89	0.45	0.739	0.98	0.60
3	100	0.993	0.97	0.95	0.58	5.151	4.45	1.74
	200	0.978	0.97	0.89	0.62	1.872	2.01	1.29
	500	0.945	0.98	0.90	0.69	0.885	0.96	0.73
	1000	0.937	0.99	0.87	0.69	0.575	0.76	0.51

Part II

Sensitivity to the design constants

The procedure has five design constants that a user must fix: the penalty weight λ_n , the inference cloud size R , the auxiliary cloud size R_{aux} , the clamping range Δ of the privacy mechanism, and the privacy budget itself. This document reports one section per constant. Validity is not at stake in any of them: the finite-sample coverage bound of Section 5 holds for every λ_n , every R_{aux} and every Δ , and depends on R only through the rank cut-off. What the sweeps measure is therefore *width* and *power*, together with the rate at which the numerical search fails.

Table 4: Scope of the sweeps. Each row is a design constant; the last column records whether the acceptance rule itself depends on it.

Constant	Grid	Model	Enters the rule?
λ_n	0, 0.1, 0.5, 1, 2, 4, 8	location-scale normal	no
R	50, 100, 200, 400, 800	location-scale normal	yes, through $a_{R,\alpha}$
R_{aux}	50, 100, 200, 400, 800	location-scale normal	no
Δ	0.5, 1, 2, 5, 10	linear regression	no, but changes the model
ε	0.1, 0.3, 1, 3	logistic regression	no, but changes the model

Not swept, and worth stating as a limitation: the finite-difference step and the direction lattice of the logistic experiment ($\text{FD_STEP} = 0.05$, $\text{DIR_GRID} = \sqrt{\lambda_n}/10$), the auxiliary cloud size in the logistic experiment (fixed at $R_{\text{aux}} = 100$ against 400 elsewhere), and the multi-start budget of the nuisance search. All three affect width rather than validity, and the last is audited separately.

4 Penalty weight λ_n

Location-scale normal, $n = 100$, $\varepsilon = 1$, $[L, U] = [0, 3]$, $\mu^* = \sigma^* = 1$, $R = 200$, $R_{\text{aux}} = 400$, $\alpha = 0.05$, 1000 replications per grid point. Setting $\lambda_n = 0$ leaves the pure nuisance-orthogonal statistic; large λ_n drives the ordering towards Mahalanobis depth. The baseline used elsewhere is $\lambda_n = 1/\log n = 0.2171$.

Table 5: Sensitivity to λ_n . Monte Carlo standard errors in parentheses; the coverage bound is 0.95025 throughout, with a Monte Carlo standard error of 0.0069. OBJ is the width objective, the average of the two widths each normalised by its grid minimum.

λ_n	μ target		σ target		Failure μ / σ	OBJ
	Coverage	Width	Coverage	Width		
0	0.994 (0.002)	2.213 (0.039)	0.989 (0.003)	0.852 (0.006)	0.001 / 0.000	2.526
0.1 *	0.980 (0.004)	0.593 (0.003)	0.976 (0.005)	0.663 (0.003)	0.010 / 0.008	1.005
0.5	0.979 (0.005)	0.587 (0.003)	0.975 (0.005)	0.688 (0.004)	0.010 / 0.010	1.019
1	0.981 (0.004)	0.605 (0.003)	0.977 (0.005)	0.712 (0.004)	0.009 / 0.011	1.052
2	0.986 (0.004)	0.624 (0.003)	0.978 (0.005)	0.746 (0.004)	0.007 / 0.011	1.093
4	0.983 (0.004)	0.642 (0.003)	0.979 (0.005)	0.774 (0.004)	0.011 / 0.011	1.130
8	0.977 (0.005)	0.657 (0.003)	0.982 (0.004)	0.791 (0.004)	0.018 / 0.015	1.156
0.1 [†]	0.971 (0.005)	0.596 (0.003)	0.979 (0.005)	0.665 (0.004)	0.011 / 0.004	

* selected. [†] independent re-evaluation of the selected value on a fresh seed stream, 1000 further replications. Two caveats. The sweep locates endpoints to a bisection tolerance of 0.1, so its absolute widths are inflated relative to the main study, which uses 10^{-3} ; the ordering in λ_n is unaffected. And the selection uses coverage of the known truth, so the chosen value is an oracle choice.

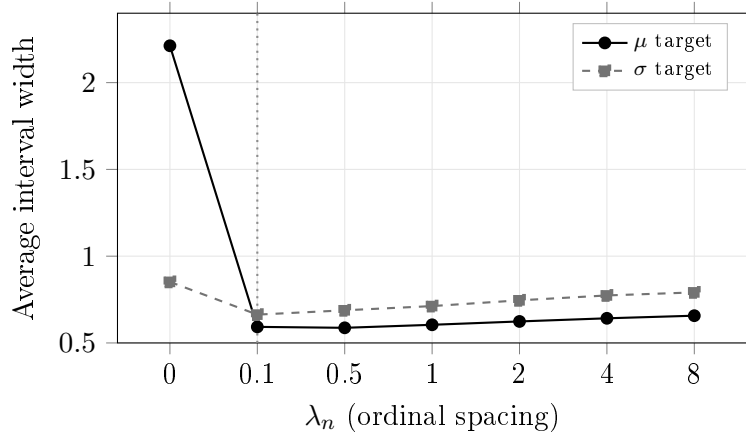


Figure 5: Sensitivity to λ_n . Grid points are at equal ordinal spacing because the design grid is geometric; the dotted vertical line marks the selected value $\lambda_n = 0.1$. The μ -target width falls from 2.213 at $\lambda_n = 0$ to 0.593 at $\lambda_n = 0.1$ and then rises slowly, so the penalty is not a cosmetic addition: with no Mahalanobis term the profile objective is flat in the nuisance direction and the acceptance set is much harder to bound. Coverage clears the Monte-Carlo-adjusted floor 0.9365 at every grid point, so the curve is about width alone.

5 Inference cloud size R

Same model and replication count. Here the rule itself moves: the cut-off is $a_{R,\alpha} = \lfloor \alpha(R+1) \rfloor + 1$ and the guaranteed coverage is $(R+2-a_{R,\alpha})/(R+1)$, which falls from 0.96078 at $R=50$ to 0.95006 at $R=800$. Coverage must therefore be read against the bound in its own row, not against a constant.

Table 6: Sensitivity to the inference cloud size R . The bound column is the guaranteed coverage at that R ; Monte Carlo standard errors in parentheses.

R	$a_{R,\alpha}$	Bound	Cov. μ	Width μ	Cov. σ	Width σ	Failure μ / σ
50	3	0.96078	0.984 (0.004)	0.642	0.986 (0.004)	0.739	0.003 / 0.002
100	6	0.95049	0.979 (0.005)	0.587	0.980 (0.004)	0.679	0.006 / 0.001
200	11	0.95025	0.981 (0.004)	0.582	0.976 (0.005)	0.669	0.005 / 0.007
400	21	0.95012	0.975 (0.005)	0.578	0.974 (0.005)	0.662	0.012 / 0.011
800	41	0.95006	0.975 (0.005)	0.574	0.976 (0.005)	0.662	0.014 / 0.008

Width falls monotonically in R , from 0.642 to 0.574 for μ , a gain of about 10% over a sixteen-fold increase in cost; the failure rate rises from 0.003 to 0.014 because a finer rank grid puts more candidates near the cut-off. $R=200$ is the point past which the bound is flat to four decimals and the width gain is under 1.5% per doubling.

6 Auxiliary cloud size R_{aux}

R_{aux} enters only the direction $C(\theta)$, never the acceptance rule, so the bound is 0.95025 in every row and the sweep is a pure cost-versus-width question.

Table 7: Sensitivity to the auxiliary cloud size R_{aux} . Monte Carlo standard errors in parentheses; the coverage bound is 0.95025 throughout.

R_{aux}	Cov. μ	Width μ	Cov. σ	Width σ	Failure μ / σ
50	0.979 (0.005)	0.582 (0.003)	0.976 (0.005)	0.668 (0.003)	0.005 / 0.008
100	0.979 (0.005)	0.582 (0.003)	0.976 (0.005)	0.669 (0.004)	0.006 / 0.005
200	0.978 (0.005)	0.582 (0.003)	0.977 (0.005)	0.670 (0.004)	0.010 / 0.006
400	0.981 (0.004)	0.582 (0.003)	0.976 (0.005)	0.669 (0.004)	0.005 / 0.007
800	0.983 (0.004)	0.581 (0.003)	0.974 (0.005)	0.669 (0.004)	0.005 / 0.005

Nothing moves. Across a sixteen-fold range the μ -target width varies between 0.5815 and 0.5825 and the σ -target width between 0.6683 and 0.6697, in both cases well inside one Monte Carlo standard error of the width itself. The direction is therefore already stable at $R_{\text{aux}}=50$ in this model, and the choice of 400 elsewhere is conservative rather than necessary. This is the one constant that can be reduced to save computation without measurable cost.

7 Clamping range Δ

Simple linear regression, $H_0 : \beta_1 = 0$, sufficient-statistic perturbation of five clamped moments under 1-GDP, $R = 200$, $\alpha = 0.05$, 1000 replications per cell. Δ changes the model, not the rule: the released moments are moments of the clamped data, so a small Δ destroys information while a large one inflates the noise. The comparison panels are the “Repro Sample” and “Parametric Bootstrap” panels of Awan and Wang (2025, Fig. 6), restricted to the cells of our design. The exact finite- R size bound is 0.04975.

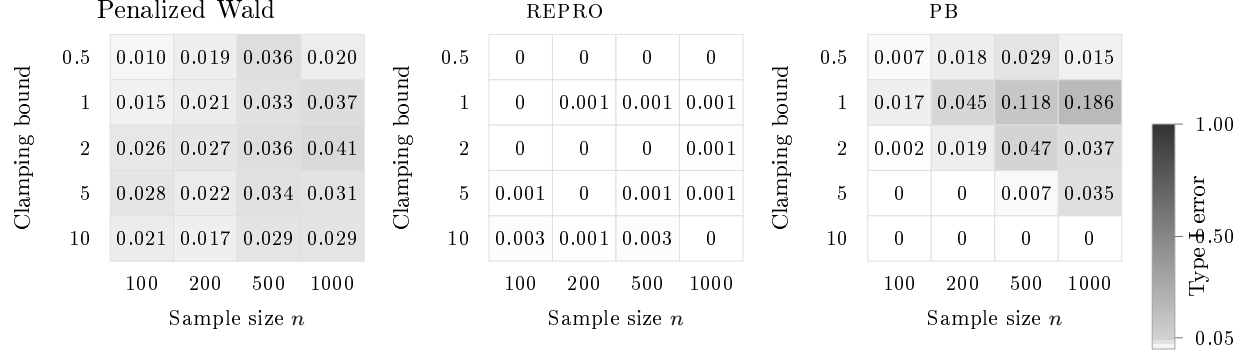


Figure 6: Empirical size at $\beta_1^* = 0$ as a function of the clamping range. The grey scale breaks at 0.05, so any dark cell is a violation of the nominal level. Ours is 0.010–0.041 over the whole grid, below the exact bound 0.04975 everywhere. REPRO is 0.000–0.003. PB reaches 0.186 at $\Delta = 1$, $n = 1000$, and the published grid extends to 0.674 (Figure 8).

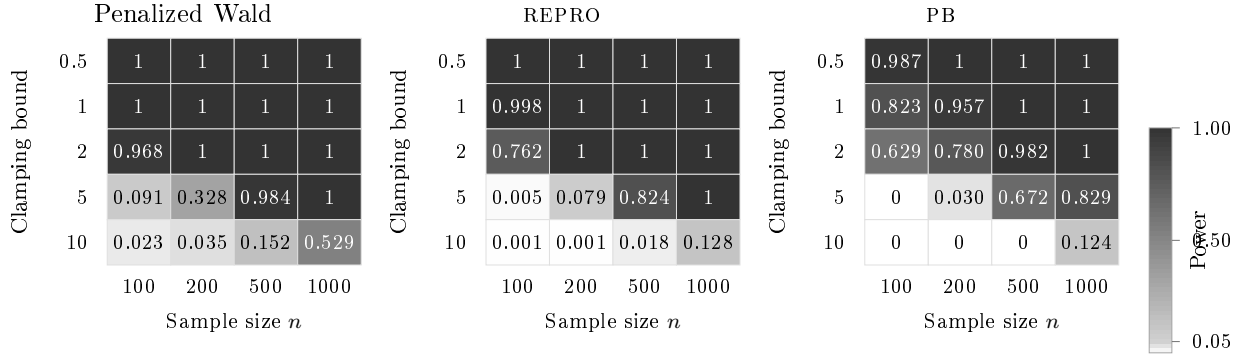


Figure 7: Power at $\beta_1^* = 1$ as a function of the clamping range. The three panels agree that $\Delta \in [0.5, 2]$ is the usable range and that power collapses by $\Delta = 10$, where the privacy noise scales as Δ^2 . Within that range our power matches REPRO at $\Delta \leq 1$ and exceeds it at $\Delta = 2$ (0.968 against 0.762 at $n = 100$) and at $\Delta = 5$ (0.091, 0.328, 0.984 against 0.005, 0.079, 0.824). PB’s apparent advantage at $\Delta = 2$, $n = 100$ cannot be read as power, since its size at that Δ is inflated.

Table 8: Clamping sweep, our runs, with Monte Carlo standard errors. All 40 cells resolved; the failure rate was zero throughout.

Δ	Size at $\beta_1^* = 0$				Power at $\beta_1^* = 1$			
	$n=100$	200	500	1000	$n=100$	200	500	1000
0.5	0.010	0.019	0.036	0.020	1.000	1.000	1.000	1.000
1	0.015	0.021	0.033	0.037	1.000	1.000	1.000	1.000
2	0.026	0.027	0.036	0.041	0.968	1.000	1.000	1.000
5	0.028	0.022	0.034	0.031	0.091	0.328	0.984	1.000
10	0.021	0.017	0.029	0.029	0.023	0.035	0.152	0.529

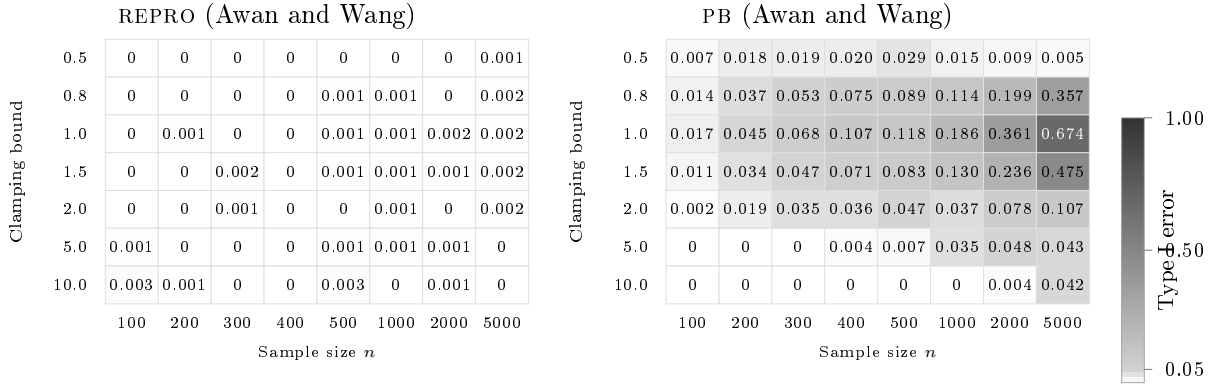


Figure 8: The published clamping grid in full, for context (Awan and Wang 2025, Fig. 6). The parametric bootstrap’s type-I error grows with the sample size rather than shrinking, reaching 0.674 at $\Delta = 1$, $n = 5000$, while the repro column stays at or below 0.003 everywhere. Our design covers the left half of these grids; the two largest sample sizes and the intermediate clamping values $\Delta \in \{0.8, 1.5\}$ were not run, and those are the cells in which the bootstrap fails hardest.

8 Privacy budget ε

Logistic regression via objective perturbation, $a^* = b^* = 1$, $\beta_0^* = 0.5$, $\beta_1^* = 2$, $R = 200$, 1000 replications per cell, all intervals at 90% confidence so the coverage bound is 0.90050. There is no separate sensitivity study for this model: the (n, ε) grid of the main experiment *is* the privacy sweep, and it is reproduced here along the ε axis. Comparison values are from Wang, Chang and Awan (Fig. 6).

Table 9: Privacy sweep for the logistic model. SAT is the number of the four sample sizes at which our interval width exceeds 18, i.e. fills all but a tenth of the search box $[-10, 10]$ and is therefore uninformative.

ε	Coverage at $n =$				Width at $n =$			
	100	200	500	1000	100	200	500	1000
<i>Penalized Wald (this paper)</i>								
0.1	0.999	0.999	1.000	0.999	19.967	19.963	19.860	18.243
0.3	1.000	0.999	0.998	0.980	19.912	19.585	13.751	3.796
1	1.000	0.997	0.983	0.946	17.366	9.658	1.648	0.739
3	0.993	0.978	0.945	0.937	5.151	1.872	0.885	0.575
<i>REPRO (Awan and Wang)</i>								
0.1	0.94	0.95	0.97	0.98	18.59	18.75	18.05	15.48
0.3	0.96	0.96	0.97	0.98	18.42	15.86	10.21	3.54
1	0.96	0.97	0.98	0.98	11.40	6.89	1.76	0.98
3	0.97	0.97	0.98	0.99	4.45	2.01	0.96	0.76
<i>PB-ADI (Wang, Chang and Awan)</i>								
0.1	0.99	0.96	0.98	0.97	9.51	10.14	11.09	8.76
0.3	0.95	0.95	0.94	0.91	6.50	6.06	3.89	1.59
1	0.91	0.94	0.89	0.89	3.15	2.37	1.06	0.60
3	0.95	0.89	0.90	0.87	1.74	1.29	0.73	0.51

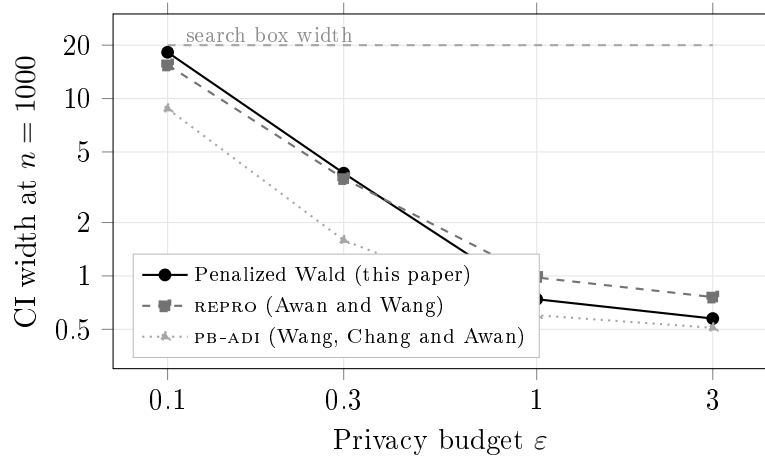


Figure 9: Interval width against the privacy budget at the largest sample size, $n = 1000$. At $\epsilon = 0.1$ our width is 18.24 and REPRO’s is 15.48 against a search box of total width 20, so both are effectively reporting the box; the comparison becomes meaningful only from $\epsilon = 0.3$ onwards, where our width falls below REPRO and the gap to PB-ADI narrows from a factor of 2.4 to a factor of 1.13. Reading relative efficiency out of the $\epsilon = 0.1$ column would be an artefact of truncation.

Part III

Behaviour at the clamping boundary

Both studies below use the clamped location-scale normal of Experiment 1: $x_i \sim N(\mu^*, \sigma^{*2})$ clamped to $[L, U] = [0, 3]$, $n = 100$, $\sigma^* = 1$, $R = 200$, $R_{\text{aux}} = 400$, $\alpha = 0.05$, and three methods per cell — Mahalanobis depth (REPRO), the Penalized Wald statistic targeting μ , and the debiased parametric bootstrap built on the adaptive indirect estimator (PB-ADI, 50 indirect samples, 200 bootstrap draws). The design constant being varied is the position of the true mean relative to the clamp: as $\mu^* \rightarrow U = 3$ an increasing share of the sample is clamped, and at $\mu^* = 3$ half of it is.

Monte Carlo resolution. These studies use 100 replications per cell, not 1000. The binomial standard error at a coverage of 0.95 is therefore 0.022, and differences smaller than about 0.04 should not be read as real. The largest effects reported below are 3 to 35 standard errors, so they survive that caveat comfortably; the smaller cell-to-cell wiggles do not.

9 Study 1: moving the mean towards the clamp

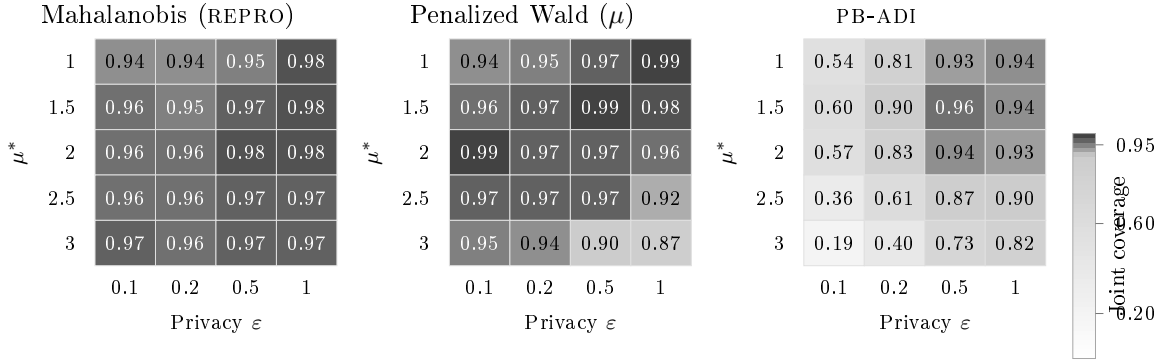


Figure 10: Study 1. Joint coverage of the nominal 95% region for (μ, σ) . Rows are the true mean μ^* , columns the privacy budget ϵ . The grey scale breaks at 0.95, so light cells are under-coverage. PB-ADI degrades in both directions at once: down the column as the mean approaches the clamp and to the left as the privacy budget shrinks, reaching 0.19 at $\mu^* = 3$, $\epsilon = 0.1$. Mahalanobis depth is flat at 0.94–0.98 over the whole grid. The Penalized Wald statistic is flat until the last row, where it drops to 0.87; Section 10 shows why.

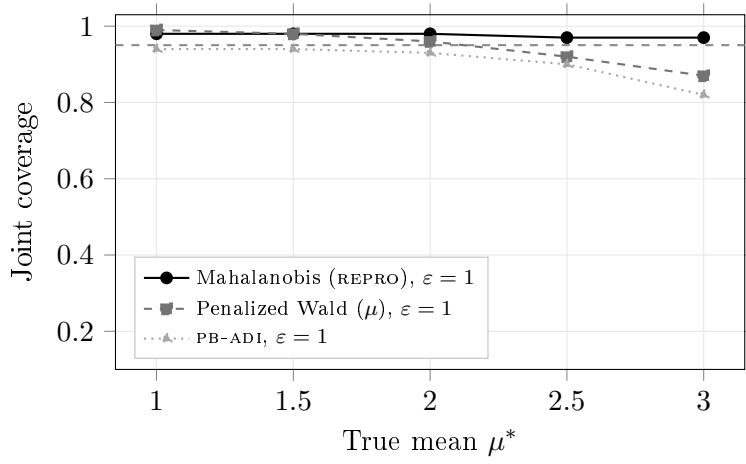


Figure 11: Study 1 at the loosest privacy budget, $\varepsilon = 1$, where the privacy noise is smallest and the clamping bias is therefore the dominant error. The dashed line is the finite- R coverage bound 0.95025. PB-ADI falls from 0.94 to 0.82 as μ^* moves from 1 to 3; the Penalized Wald statistic falls from 0.99 to 0.87; Mahalanobis depth does not move. That the two repro-family methods separate is the point: the loss is not a property of the framework but of the statistic's dependence on the Jacobian of the clamped mean.

Table 10: Study 1 in full. Joint and marginal coverage of the nominal 95% region, and average width of the marginal interval for μ . The coverage bound is 0.95025 and the Monte Carlo standard error at that level is 0.022 on 100 replications.

μ^*	ε	Joint coverage			Coverage of μ			Width for μ		
		REPRO	PW	PB-ADI	REPRO	PW	PB-ADI	REPRO	PW	PB-ADI
1	0.1	0.94	0.94	0.54	0.98	0.97	0.82	4.313	3.972	3.988
	0.2	0.94	0.95	0.81	0.99	0.95	0.93	2.788	1.960	2.157
	0.5	0.95	0.97	0.93	0.99	0.99	0.95	0.928	0.758	0.657
	1	0.98	0.99	0.94	1.00	0.99	0.96	0.673	0.568	0.467
1.5	0.1	0.96	0.96	0.60	0.98	0.96	0.92	4.892	4.417	5.317
	0.2	0.95	0.97	0.90	0.99	0.99	0.99	2.385	1.649	1.750
	0.5	0.97	0.99	0.96	0.99	0.99	0.97	0.756	0.633	0.531
	1	0.98	0.98	0.94	1.00	0.99	0.97	0.610	0.522	0.437
2	0.1	0.96	0.99	0.57	1.00	0.99	0.91	4.301	3.932	6.825
	0.2	0.96	0.97	0.83	0.99	0.98	0.97	2.817	2.039	3.299
	0.5	0.98	0.97	0.94	0.99	0.99	0.95	0.992	0.806	0.677
	1	0.98	0.96	0.93	0.99	0.99	0.97	0.691	0.590	0.478
2.5	0.1	0.96	0.97	0.36	1.00	0.98	0.74	3.467	3.097	7.712
	0.2	0.96	0.97	0.61	1.00	0.98	0.79	2.757	2.503	5.769
	0.5	0.97	0.97	0.87	1.00	0.98	0.88	1.831	1.365	1.677
	1	0.97	0.92	0.90	1.00	0.96	0.93	0.968	0.795	0.678
3	0.1	0.97	0.95	0.19	1.00	0.96	0.55	3.090	2.574	7.823
	0.2	0.96	0.94	0.40	1.00	0.95	0.49	2.456	2.295	7.060
	0.5	0.97	0.90	0.73	1.00	0.93	0.73	2.176	1.729	4.471
	1	0.97	0.87	0.82	1.00	0.88	0.82	1.770	1.443	1.771

Note on the width columns: the search box for σ is $[0.1, 5]$, so a σ -width of 4.9 is the whole box. At $\varepsilon = 0.1$ the σ -widths are 4.65–4.90 for both repro-family methods, i.e. box-saturated, so the $\varepsilon = 0.1$ column carries no information about efficiency in the scale parameter. PB-ADI's μ -widths at $\mu^* \geq 2$ are the largest of the three despite its being the narrowest method at $\mu^* = 1$: the indirect estimator is being pushed away from the truth by the clamping bias, and the bootstrap inherits the resulting spread.

10 Study 2: hard versus soft clamping at $\mu^* = 3$

Fix $\mu^* = 3$, the worst cell of Study 1, and replace the hard clamp $[x]_L^U$ by the logistic map

$$T(x) = L + \frac{U - L}{1 + \exp(-k(x - \frac{L+U}{2}))}, \quad k = 1.5,$$

which sends $\mathbb{R}$ strictly into (L, U) . Because the transformed data still lie in $[L, U]$, the sensitivities $(U - L)/n$ and $(U - L)^2/n$ are unchanged and the noise calibration is identical: the two regimes are compared at exactly the same privacy cost. The clamp type is threaded through the release, the region grid, the indirect estimator and the bootstrap resampler, so no method is solved under a mismatched model.

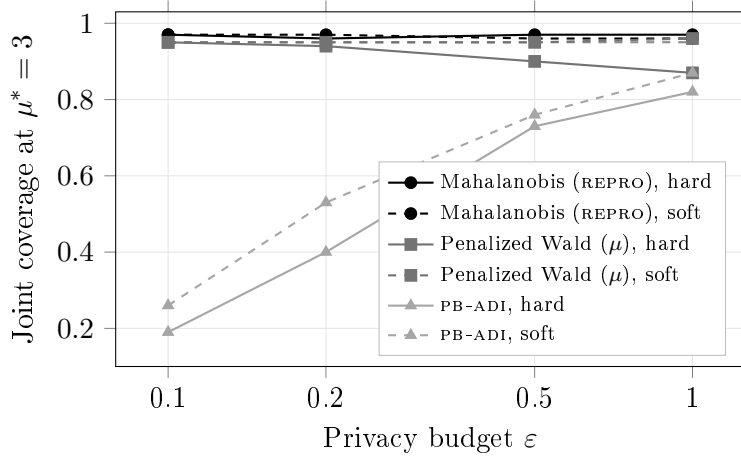


Figure 12: Study 2. Joint coverage at $\mu^* = 3$ under hard (solid) and soft (dashed) clamping, at identical privacy cost. The dashed horizontal line is the finite- R bound 0.95025. Softening the clamp restores the Penalized Wald statistic to 0.95–0.96 from 0.87–0.95, with the largest repair exactly where the hard-clamp loss was largest. PB-ADI improves but is not repaired: $0.19 \rightarrow 0.26$, $0.40 \rightarrow 0.53$, $0.73 \rightarrow 0.76$, $0.82 \rightarrow 0.87$. Mahalanobis depth is unchanged, as it must be, since it never differentiates the mechanism.

Table 11: Study 2 in full, $\mu^* = 3$, $\sigma^* = 1$, 100 replications per cell. Widths are for the marginal interval for μ ; areas are of the joint region.

Method	ε	Joint coverage		Coverage of μ		Width for μ		Area	
		hard	soft	hard	soft	hard	soft	hard	soft
Mahalanobis (repro)	0.1	0.97	0.97	1.00	1.00	3.090	3.312	10.75	11.96
	0.2	0.96	0.97	1.00	1.00	2.456	2.800	5.62	6.71
	0.5	0.97	0.96	1.00	1.00	2.176	2.205	2.45	2.52
	1	0.97	0.96	1.00	1.00	1.770	1.412	1.33	0.97
Penalized Wald (μ)	0.1	0.95	0.95	0.96	0.97	2.574	3.230	11.71	14.68
	0.2	0.94	0.95	0.95	0.97	2.295	2.757	9.32	11.16
	0.5	0.90	0.95	0.93	0.97	1.729	1.958	4.26	4.98
	1	0.87	0.96	0.88	0.98	1.443	1.154	2.16	1.78
PB-ADI	0.1	0.19	0.26	0.55	0.55	7.823	7.842	31.37	32.50
	0.2	0.40	0.53	0.49	0.61	7.060	6.809	26.61	23.53
	0.5	0.73	0.76	0.73	0.78	4.471	2.943	12.97	7.60
	1	0.82	0.87	0.82	0.88	1.771	1.010	3.37	1.35

11 Reading

Three conclusions, in decreasing order of how firmly the data support them.

The debiased parametric bootstrap fails at the clamping boundary, and strong privacy makes it worse. Joint coverage of the nominal 95% region falls to 0.19 at $\mu^* = 3$, $\varepsilon = 0.1$, and is below 0.75 in eight of the twenty cells. The failure is not a small-sample artefact of the bootstrap draw count: it is monotone in μ^* at every ε and monotone in ε at every μ^* . The mechanism is the one the debiased-bootstrap literature identifies — the estimator is biased by clamping and the bootstrap inherits the bias — except that the adaptive indirect estimator is supposed to remove exactly that bias. What these cells show is that its bias correction is itself degraded when the clamp is active for a large fraction of the sample.

Mahalanobis depth is insensitive to the boundary. Joint coverage stays in $[0.94, 0.98]$ across all twenty cells, and marginal coverage for μ is 0.98–1.00. This is the expected behaviour: the statistic is a function of the pooled cloud alone and never differentiates the mechanism, so a degenerate Jacobian cannot reach it. The price is the over-coverage and excess width documented elsewhere.

The Penalized Wald statistic partially inherits the problem, and the soft clamp identifies the cause. At $\mu^* = 3$ its joint coverage falls to 0.87 at $\varepsilon = 1$ and its marginal coverage for μ to 0.88, which is 3.2 standard errors below the bound and the only under-coverage we observe for the method anywhere in this paper. Its direction $C(\theta) = \Sigma^{-1}[J_\beta - J_\eta(J_\eta^\top \Sigma^{-1} J_\eta)^{-1} J_\eta^\top \Sigma^{-1} J_\beta]$ depends on the Jacobian of the mean release, and with a hard clamp that Jacobian is a mean of indicator-weighted terms whose informative part vanishes as the clamp becomes active. Replacing the clamp by a differentiable map with nowhere-zero derivative, at identical privacy cost, returns coverage to 0.95–0.96 and marginal coverage to 0.97–0.98. That is a controlled experiment on the mechanism rather than on the method, and it isolates the cause to the non-differentiability of the hard clamp rather than to the orthogonalisation itself.

A practical consequence worth stating: when a mechanism is known to clamp a substantial fraction of the sample, a direction built from a finite-difference Jacobian of that mechanism should be treated with suspicion, and either a smoothed clamp or a statistic that does not use the Jacobian is preferable. The soft clamp costs nothing in privacy and, in these runs, nothing in width either — at $\varepsilon = 1$ it is narrower for every method.

Part IV

Provenance

12 Where each number comes from

Nothing in this document is read from a file at compile time; every value is written into the source. The values divide into two kinds, and the captions say which is which wherever both appear in one figure.

Our own runs

Quantity	Produced by
Experiment 1: coverage, width, area, targeted width reduction	<code>exp1_location_scale_normal/exp1_coverage.R</code> , summarised in <code>results_coverage/coverage_study_n100_ep1_nSIM1000.csv</code> and <code>coverage_targeted_efficient.csv</code>
Experiment 1: λ_n , R_{aux} and R sweeps	<code>sensitivity_analysis/exp1_normal_sensitivity/</code> , files <code>lambda_grid_tuning_results.csv</code> , <code>R_aux_sensitivity_results.csv</code> , <code>R_point_cloud_sensitivity_results.csv</code>
Experiment 2: size and power grid	<code>exp2_linear_regression/linearreg_eff.R</code> , summarised in <code>results_eff/Efficient_INT.csv</code>
Experiment 2: clamping sweep	<code>sensitivity_analysis/exp2_linear_regression_sensitivity/</code> , file <code>exp2_delta_sensitivity_long.csv</code>
Experiment 3: coverage and width	<code>exp3_objective_perturbation/eff.R</code> , aggregated in <code>results/heatmap_data.csv</code>
Clamping-boundary studies	<code>PB_adi_failure/failure_v1.R</code> and <code>failure_v2.R</code> , summarised in <code>master_grid_summary.csv</code> and <code>master_softclamp_summary.csv</code>

Published comparison values

Panel	Source
REPRO, Experiment 2 size and power	Awan and Wang (2025), Figure 5, panel “Repro Sample”
PB-ADI, Experiment 2	Wang, Chang and Awan, Figure 5, panel “PB (Indirect estimator + approximate-pivot)”
PB-NAIVE, Experiment 2	Wang, Chang and Awan, Figure 5, panel “PB (Naive estimator + F -statistic)”
REPRO and PB, clamping sweep	Awan and Wang (2025), Figure 6, all four panels
REPRO, PB-ADI and PB-NAIVE, Experiment 3	Wang, Chang and Awan, Figure 6, panels “Repro Samples”, “Debiased Estimator” and “Naive Estimator”

Published grids are shown only on the cells of our own design; the full published clamping grid is reproduced once, in Figure 8, for context. Where a published grid does not contain a cell of our design, the cell is left blank rather than dropped.

Reference levels used throughout

Study	α	R	cut-off $a_{R,\alpha}$	size bound	coverage bound
Experiments 1 and 2, and their sweeps	0.05	200	11	0.04975	0.95025
Experiment 1, R sweep at $R = 50$	0.05	50	3	0.03922	0.96078
Experiment 1, R sweep at $R = 100$	0.05	100	6	0.04950	0.95050
Experiment 1, R sweep at $R = 400$	0.05	400	21	0.04988	0.95012
Experiment 1, R sweep at $R = 800$	0.05	800	41	0.04994	0.95006
Experiment 3	0.10	200	21	0.09950	0.90050

The cut-off is $a_{R,\alpha} = \lfloor \alpha(R+1) \rfloor + 1$, the size bound $(a_{R,\alpha} - 1)/(R+1)$ and the coverage bound its complement. The bound is an upper bound on size and a lower bound on coverage; it is attained when ties in the depth have probability zero. Monte Carlo standard errors at the coverage bound are 0.0069 at 1000 replications, 0.0095 for Experiment 3, and 0.0217 at the 100 replications of the clamping-boundary studies.